

649 A.1 Additional Results

650 We provide a detailed breakdown of the evaluation metrics. For streaming models with single-frame output ($k = 1$), we
 651 report the decomposed COCOLA harmonic score and COCOLA percussive score across different future visibility t_f in
 652 Fig. 6. For streaming models with combinations of output chunk size $k > 1$ and future visibility t_f , we present the results
 653 of other metrics: Fig. 7 displays the COCOLA harmonic and percussive scores, while Fig. 8 reports the rhythmic alignment
 654 (Beat F_1) and Audio Quality (FAD) scores.

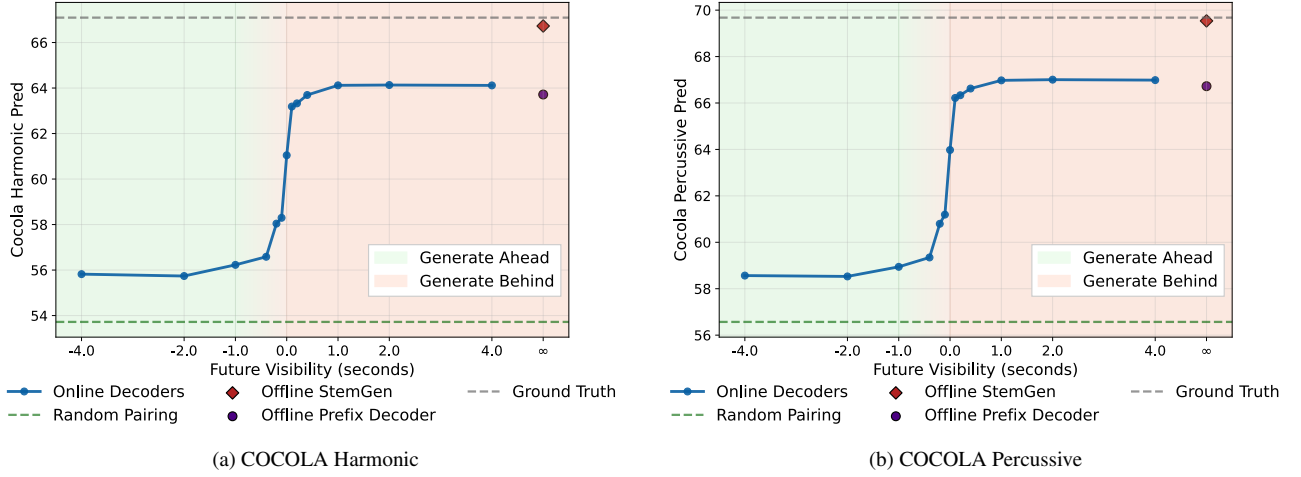


Figure 6. COCOLA harmonic score and COCOLA percussive score of generated accompaniment for streaming models with $k = 1$ across different t_f .

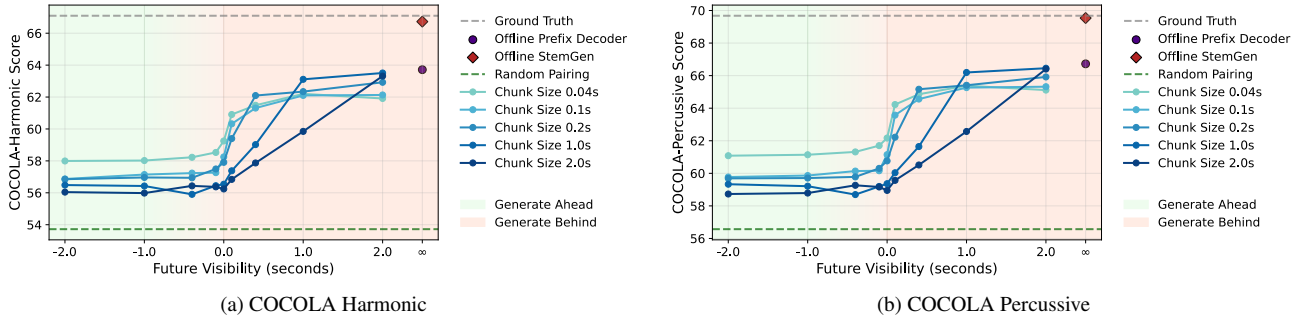


Figure 7. COCOLA harmonic score and COCOLA percussive score of generated accompaniment for streaming models with combinations of $k > 1$ and t_f .

655 A.2 Transformer Backbone

656 We use a transformer backbone similar to Llama 3. Following current large-model practice, the network uses pre layer
 657 normalization throughout [41], with Root Mean Square Normalization as the normalization operator [42]; we adopt the
 658 simplified RMSNorm variant reported to work well at large scale [43]. Positional information is injected by rotary position
 659 embedding in self-attention [44]. Inside attention, we apply query-key normalization by normalizing queries and keys
 660 along the head dimension before the similarity computation [45], together with a dimension-dependent scaling factor on
 661 the normalized scores as recommended by recent scaling results [46]. The feed-forward sublayers use the gated SwiGLU
 662 activation [47]. To improve decoding efficiency while preserving quality, key-value projections are shared across groups
 663 of query heads, i.e., grouped-query attention [48]. The decoder also supports cross-attention for conditioning on external
 664 context.

665 A.3 Listening Test Details

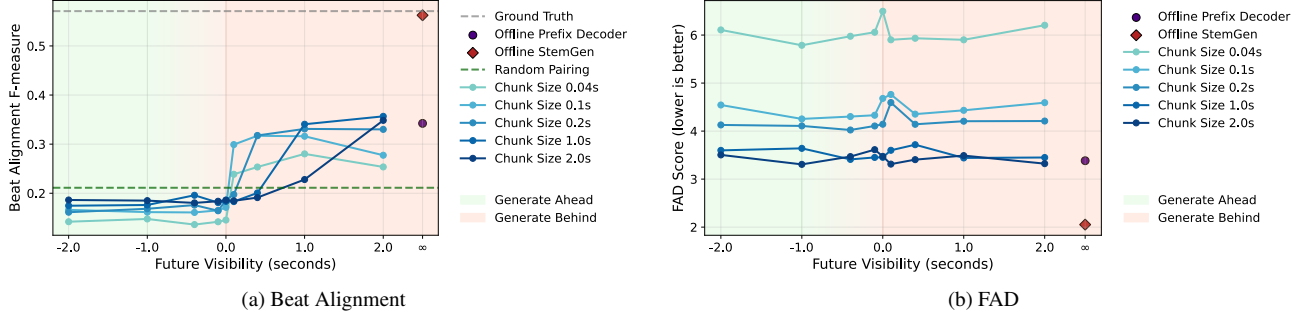


Figure 8. Beat alignment and FAD score of generated accompaniment for streaming models with combinations of $k > 1$ and t_f .

	Ground Truth	Offline StemGen	Offline Prefix Decoder	$t_f = 1$	$t_f = 0$	$t_f = -1$	Random Pairing
Ground Truth	N/A	!	*	*	*	*	*
Offline StemGen	!	N/A	!	!	!	*	*
Offline Prefix Decoder	*	!	N/A	!	!	*	!
$t_f = 1$	*	!	!	N/A	!	*	!
$t_f = 0$	*	!	!	!	N/A	*	!
$t_f = -1$	*	*	*	*	*	N/A	!
Random Pairing	*	*	!	!	!	!	N/A

Table 1. Pairwise statistical significance results for listening study. * indicates significant difference ($p < 0.05/21$), ! indicates non-significant ($p > 0.05/21$).

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